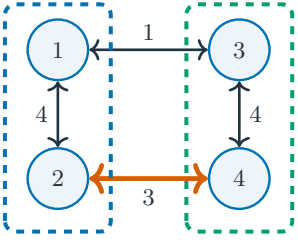
1. The process



Predict the mass in A.
Rates: inverse time units.

2. What is lost?

Both starting states give
$p_A = 1, \quad p_B = 0$

Start in 1: $p_A'(0) = -1$

Start in 2: $p_A'(0) = -3$

Same coarse state, different futures.

3. Recover the future

Keep one imbalance z

$$y' = -4y + 2z$$
$$z' = 2y - 12z$$

↓

Or eliminate z and retain its history:

$$y' = -4y + 2e^{-12t}z(0) + 4\int_0^t e^{-12(t-s)}y(s)\,ds$$